PATCHA MANIKANTA NAGA VENKATA SAI GANESH

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OBJECTIVE

Computer Science and Data Science undergraduate with hands-on experience in full-stack development, Generative AI, and deep learning. Skilled in building RESTful APIs, AI-powered applications, and machine learning prototypes. Seeking an entry-level software development role to build scalable solutions and contribute to real-world applications.

PROFESSIONAL EXPERIENCE

MERN Stack Developer Intern | PurpleLane **February 22, 2026 – August 22, 2026**

Developed and enhanced full-stack web application features using React.js, Node.js, Express.js, and MongoDB. Built reusable UI components, integrated RESTful APIs, implemented backend functionality, and managed database operations. Focused on responsive design, modular code development, and application performance while following maintainable software development practices.

ACADEMIC QUALIFICATION

B.Tech — Computer Science and Engineering (Data Science) **Expected Graduation: 2027**
Sasi Institute of Technology and Engineering **CGPA: 8.34/10**

Intermediate **2023**
Aditya Junior College **Percentage: 70%**

SSC **2021**
St. Ann's E.M. School **Percentage: 92%**

TECHNICAL SKILLS

Programming: JavaScript, Python
Frontend: React.js, HTML5, CSS3, Tailwind CSS, Vite
Backend: Node.js, Express.js, RESTful APIs
Databases: MongoDB, PostgreSQL
Generative AI: LangChain, LangGraph, RAG, LLMs, AI Agents, Prompt Engineering, Google Gemini API, Pinecone
Tools and Platforms: Git, GitHub, Postman, VS Code, Docker, Vercel, Render

PROJECTS

Student Intelligence — AI-Powered Student Assistant
Technologies: React, Vite, Node.js, Express.js, Ollama, Llama 3.2, JavaScript, REST APIs
- Integrated the locally running Llama 3.2 model through Ollama to enable AI-powered student conversations.
- Developed backend chat and health-monitoring APIs with a dedicated service for local model communication.
- Implemented model detection and fallback handling to improve application reliability.

ShopMate — AI-Powered E-Commerce Platform
Technologies: React.js, Vite, Tailwind CSS, Node.js, Express.js, MongoDB Native Driver, JWT, Pinecone, Google Gemini AI, LangChain, LangGraph
- Developed a full-stack e-commerce platform with authentication, product management, and keyword-based search.
- Integrated semantic product search, AI-generated product descriptions, and multimodal image-based product listing.
- Built AI customer support, a RAG-based policy assistant, and an AI blog-generation workflow with human review.

NEPTUNE-CXR — Deep Learning-Based Chest X-Ray Disease Probability Detection
Technologies: Python, PyTorch, DenseNet121, FastAPI, React.js, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Grad-CAM
- Developed a deep learning prototype estimating probabilities across 14 thoracic disease categories from chest X-ray images.
- Implemented image preprocessing, probability visualization, Grad-CAM heatmaps, report storage, and an analytics dashboard.

LegalGenie — AI-Powered Legal Research Assistant
Technologies: React.js, Node.js, Express.js, MongoDB, LangChain, ChromaDB, Gemini AI, REST APIs
- Developed semantic search and RAG-based legal research for retrieving relevant cases and generating contextual answers.
- Implemented PDF ingestion, document processing, vector retrieval, and AI-generated case summaries and briefs.

Order Management System
Technologies: React.js, Node.js, Express.js, MongoDB, Material UI, CSS, REST APIs
- Developed a full-stack application for product, customer, inventory, and order management.
- Implemented CRUD operations, RESTful APIs, order status tracking, database integration, and error handling.

CERTIFICATES

- MERN Stack — PurpleLane
- Python 101 for Data Science — IBM SkillsBuild
- Generative AI — PurpleLane

ADDITIONAL INFORMATION

Achievement: Participated in SIH Internal Hackathon 2026. | **Membership:** IEEE Member.
Languages: English, Telugu, Hindi.